

# Sampling Distributions and Hypothesis Tests in the Linear Regression Model

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# Chapter Outline

- The sampling distribution of the OLS estimator.
- The standard error of  $\hat{\beta}_1$ .
- Hypothesis tests concerning  $\beta_1$ .
- Confidence intervals for  $\beta_1$ .
- Regression when  $X$  is binary.
- Heteroskedasticity and homoskedasticity.
- Efficiency of *OLS* and the *Student t* distribution.

# A Big Picture Review Of Where We Are Going

We want to learn about the slope of the population regression line. We have data from a sample, so there is sampling uncertainty. There are five steps towards this goal:

- 1 State the population object of interest.
- 2 Provide an estimator of this population object.
- 3 Derive the sampling distribution of the estimator (this requires certain assumptions). In large samples this sampling distribution will be normal by the *CLT*.
- 4 The square root of the estimated variance of the sampling distribution is the standard error (*SE*) of the estimator.
- 5 Use the *SE* to construct t-statistics (for hypothesis tests) and confidence intervals.

# The sampling distribution of $\hat{\beta}_1$

Like  $\bar{Y}$ ,  $\hat{\beta}_1$  has a sampling distribution.

- What is  $E(\hat{\beta}_1)$ —If  $E(\hat{\beta}_1) = \beta_1$ , then OLS is unbiased, which is a good thing.
- What is  $\text{Var}(\hat{\beta}_1)$ ? (measure of sampling uncertainty)—We need to derive a formula so that we can compute the standard error of  $\hat{\beta}_1$ .
- What is the distribution of  $\hat{\beta}_1$  in small samples? It is very complicated in general.
- What is the distribution of  $\hat{\beta}_1$  in large samples? IN large samples,  $\hat{\beta}_1$  is normally distributed.

# The mean and variance of the sampling distribution of $\hat{\beta}_1$

Some preliminary algebra:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

$$\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{u}$$

$$Y_i - \bar{Y} = \beta_1(X_i - \bar{X}) + (u_i - \bar{u})$$

Thus,

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X})(\beta_1(X_i - \bar{X}) + (u_i - \bar{u}))}{\sum_{i=1}^n (X_i - \bar{X})^2}\end{aligned}$$

## The mean and variance of the sampling distribution of $\hat{\beta}_1$

$$\hat{\beta}_1 = \beta_1 \frac{\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\text{so } \hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Now,

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u}) &= \sum_{i=1}^n (X_i - \bar{X})u_i - \left[ \sum_{i=1}^n (X_i - \bar{X}) \right] \bar{u} \\ &= \sum_{i=1}^n (X_i - \bar{X})u_i - \left[ \left( \sum_{i=1}^n X_i \right) - n\bar{X} \right] \bar{u} \\ &= \sum_{i=1}^n (X_i - \bar{X})u_i \end{aligned}$$

## The mean and variance of the sampling distribution of $\hat{\beta}_1$

Substitute  $\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u}) = \sum_{i=1}^n (X_i - \bar{X})u_i$  into the expression for  $\hat{\beta}_1 - \beta_1$ :

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

OLS intercept in terms of errors  $u_i$

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})u_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

## The mean and variance of the sampling distribution of $\hat{\beta}_1$

$$\begin{aligned} E(\hat{\beta}_1) - \beta_1 &= E \left[ \frac{\sum_{i=1}^n (X_i - \bar{X}) u_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \right] \\ &= E \left\{ E \left[ \frac{\sum_{i=1}^n (X_i - \bar{X}) u_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \mid X_1, \dots, X_n \right] \right\} \\ &= E \left\{ \frac{\sum_{i=1}^n (X_i - \bar{X}) E(u_i \mid X_i)}{\sum_{i=1}^n (X_i - \bar{X})^2} \right\} \\ &= 0 \quad \text{because } E(u_i \mid X_i = x) = 0 \text{ by the OLS A.1} \end{aligned}$$

- Thus, OLS A.1 implies that  $E(\hat{\beta}_1) = \beta_1$ .
- That is,  $\hat{\beta}_1$  is an **unbiased estimator** of  $\beta_1$ .

## The mean and variance of the sampling distribution of $\hat{\beta}_1$

Next we calculate  $\text{Var}(\hat{\beta}_1)$ : write

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X}) u_i}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\left(\frac{n-1}{n}\right) s_X^2}$$

where  $v_i = (X_i - \bar{X}) u_i$ . If  $n$  is large,  $s_X^2 \xrightarrow{P} \sigma_X^2$  and  $\frac{n-1}{n} \rightarrow 1$ , so

$$\hat{\beta}_1 - \beta_1 \xrightarrow{P} \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\sigma_X^2},$$

# The mean and variance of the sampling distribution of $\hat{\beta}_1$

Thus,

$$\hat{\beta}_1 - \beta_1 \xrightarrow{P} \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\sigma_X^2},$$

so  $\text{Var}(\hat{\beta}_1 - \beta_1) = \text{Var}(\hat{\beta}_1)$ . Thus,

$$\text{Var}(\hat{\beta}_1) = \frac{\text{Var}\left(\frac{1}{n} \sum_{i=1}^n v_i\right)}{(\sigma_X^2)^2} = \frac{\text{Var}(v_i)/n}{(\sigma_X^2)^2}$$

where the final equality uses OLS A.2. Notice that  $v_i \xrightarrow{P} (X_i - \mu_x)u_i$  hence

## Variance of $\hat{\beta}_1$

$$\text{Var}(\hat{\beta}_1) = \frac{1}{n} \times \frac{\text{Var}((X_i - \mu_x)u_i)}{(\sigma_X^2)^2}$$

Summary so far:

- $\hat{\beta}_1$  is unbiased:  $E(\hat{\beta}_1) = \beta_1$  just like  $\bar{Y}$ .
- $\text{Var}(\hat{\beta}_1)$  is inversely proportional to  $n$ —just like  $\bar{Y}$ .

## What is the sampling distribution of $\hat{\beta}_1$ ?

The exact sampling distribution is complicated—it depends on the population distribution of  $(Y_i, X_i)_{i=1}^n$ , but when  $n$  is large we get some simple and good approximations.

- 1 Because  $\text{Var}(\hat{\beta}_1) \propto \frac{1}{n}$  and  $E(\hat{\beta}_1) = \beta_1$ ,  $\hat{\beta}_1 \xrightarrow{P} \beta_1$
- 2 When  $n$  is large, the sampling distribution of  $\hat{\beta}_1$  is well approximated by a normal distribution (CLT)

### The CLT in this context

Suppose  $v_i$ ,  $i = 1, \dots, n$  is i.i.d. with  $E(v) = 0$  and  $\text{Var}(v_i) = \sigma_v^2$ . Then, when  $n$  is large,  $\frac{1}{n} \sum_{i=1}^n v_i$  is approximately distributed  $N(0, \sigma_v^2/n)$ .

## Large sample approximation to the distribution of $\hat{\beta}_1$

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\left(\frac{n-1}{n}\right) s_X^2} \xrightarrow{P} \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\sigma_X^2}$$

where  $v_i = (X_i - \bar{X})u_i$

- When  $n$  is large,  $v_i \xrightarrow{P} (X_i - \mu_X)u_i$ , which is i.i.d. (OLS A.2) and  $\text{Var}(v_i) < \infty$  (OLS A.3).
- So by the CLT,  $\frac{1}{n} \sum_{i=1}^n v_i$  is approximately distributed  $N(0, \sigma_v^2/n)$ .
- Thus, for  $n$  large  $\hat{\beta}_1$  is approximately distributed:

### Large Sample distribution of $\hat{\beta}_1$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma_v^2}{n(\sigma_X^2)^2}\right)$$

where  $\sigma_v^2 = \text{Var}((X_i - \mu_X)u_i)$ .

- The larger the variance of  $X$ , the smaller the variance of  $\hat{\beta}_1$ .

# Discussion

## The math

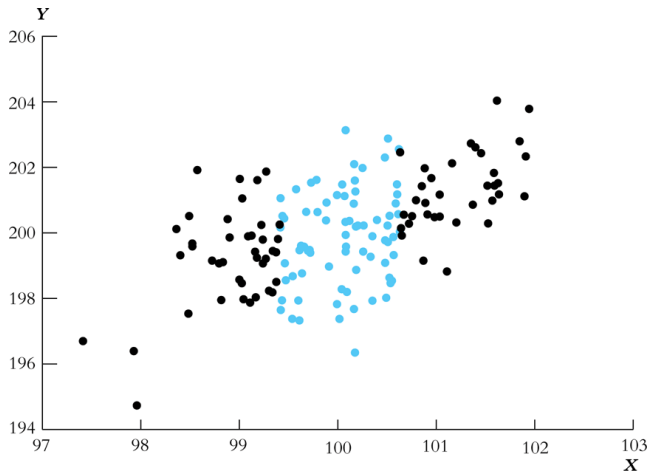
$$\text{Var}(\hat{\beta}_1 - \beta_1) = \frac{1}{n} \times \frac{\text{Var}((X_i - \mu_X)u_i)}{(\sigma_X^2)^2}$$

where  $\sigma_X^2 = \text{Var}(X_i)$ . The variance of  $X$  appears (squared) in the denominator—so increasing the spread of  $X$  decreases the variance of  $\hat{\beta}_1$ .

## The intuition

If there is more variation in  $X$ , then there is more information in the data that you can use to fit the regression line. This is most easily seen in a figure.

The larger the variance of  $X$ , the smaller the variance of  $\hat{\beta}_1$



The number of black and blue dots is the same. Using which would you get a more accurate regression line?

## Summary of the sampling distribution of $\hat{\beta}_1$

If the three Least Squares Assumptions hold, then

- The exact (finite sample) sampling distribution of  $\hat{\beta}_1$  has:
  - $E(\hat{\beta}_1) = \beta_1$  (that is,  $\hat{\beta}_1$  is unbiased)
  - $\text{Var}(\hat{\beta}_1) = \frac{1}{n} \times \frac{(X_i - \mu_X)u_i}{(\sigma_X^2)^2} \propto \frac{1}{n}$ .
- Other than its mean and variance, the exact distribution of  $\hat{\beta}_1$  is complicated and depends on the distribution of  $(X, u)$
- $\hat{\beta}_1 \xrightarrow{P} \beta_1$  (that is,  $\hat{\beta}_1$  is consistent)
- When  $n$  is large,  $\frac{\hat{\beta}_1 - E(\hat{\beta}_1)}{\sqrt{\text{Var}(\hat{\beta}_1)}} \sim N(0, 1)$  (CLT)
- Very similar results to the sampling distribution of  $\bar{Y}$ .

# Sampling Distribution of the *OLS* Estimator

## Large Sample Distribution of $\hat{\beta}_1$ and $\hat{\beta}_0$ .

If the three *OLS* assumptions hold and the sample size  $n$  is large,  $\hat{\beta}_1$  and  $\hat{\beta}_0$  are jointly normally distributed. The distribution of  $\hat{\beta}_1$  is  $N(\beta_1, \sigma_{\hat{\beta}_1}^2)$  where

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\text{Var}[(X_i - \mu_X)u_i]}{(\text{Var}[X_i])^2}. \quad (1)$$

The distribution of  $\hat{\beta}_0$  is  $N(\beta_0, \sigma_{\hat{\beta}_0}^2)$  where

$$\sigma_{\hat{\beta}_0}^2 = \frac{1}{n} \frac{\text{Var}[H_i u_i]}{(E[H_i^2])^2}, \quad \text{where } H_i = 1 - \frac{\mu_X}{E[X_i^2]} X_i. \quad (2)$$

# Testing Hypothesis About One Of The Regression Coefficients

- **Object of interest:**  $\beta_1$  in  $Y_i = \beta_0 + \beta_1 X_i + u_i$ .
- **Estimator:** *OLS* estimator  $\hat{\beta}_1$ .
- **Sampling distribution of  $\hat{\beta}_1$ :** To derive the large sample distribution of  $\hat{\beta}_1$  we make the following assumptions
  - $E[u_i|X_i] = 0$ .
  - $(X_i, Y_i)$  for  $i = 1, \dots, n$  are i.i.d.
  - Large outliers are rare  $E[X^4] < \infty$  and  $E[Y^4] < \infty$ .

# Testing Hypothesis About One Of The Regression Coefficients

## The Sampling Distribution of the *OLS* Estimator

Under the least squares assumptions, for  $n$  large,  $\hat{\beta}_1$  is distributed as

$$\hat{\beta}_1 \sim N(\beta_1, \sigma_{\hat{\beta}_1}^2), \quad (3)$$

where

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\sigma_{v_i}^2}{(\sigma_X^2)^2}$$

and

$$v_i = (X_i - \mu_X) u_i.$$

# Testing Hypothesis About One Of The Regression Coefficients

## General Setup

The objective is to test a hypothesis, like  $\beta_1 = 0$ , using data –to reach a tentative conclusion whether the (null) hypothesis is correct or incorrect.

- Null hypothesis and two-sided alternative

$$H_0 : \beta_1 = \beta_{1,0} \text{ vs. } H_1 : \beta_1 \neq \beta_{1,0}, \quad (4)$$

- Null hypothesis and one-sided alternative

$$H_0 : \beta_1 = \beta_{1,0} \text{ vs. } H_1 : \beta_1 < \beta_{1,0}, \quad (5)$$

where  $\beta_{1,0}$  is the hypothesized value under the null.

# Testing Hypothesis About One Of The Regression Coefficients

## General Approach

Construct a t-statistic and compute the p-value (or compare the t-statistic to the  $N(0, 1)$  critical value).

- **In general:**  $t = \frac{\text{estimator} - \text{hypothesized value}}{\text{standard error of the estimator}}.$
- **For testing about  $\mu$  we had:**  $t = \frac{\bar{Y} - \mu_{Y,0}}{\frac{s_Y}{\sqrt{n}}}.$
- **For testing about  $\beta_1$  we have:**  $t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)},$

where  $SE(\hat{\beta}_1)$  is the square root of an estimator of the variance of the sampling distribution of  $\hat{\beta}_1$ .

# Testing Hypothesis About One Of The Regression Coefficients

## Standard Error of the *OLS* estimator

Recall the expression for the variance of  $\hat{\beta}_1$  when the sample size is large

$$\sigma_{\hat{\beta}_1}^2 = \text{Var}[\hat{\beta}_1] = \frac{1}{n} \frac{\text{Var}[(X_i - \mu_X)u_i]}{(\sigma_X^2)^2} = \frac{1}{n} \frac{\sigma_{v_i}^2}{(\sigma_X^2)^2}, \quad (6)$$

where  $v_i = (X_i - \mu_X)u_i$ .

Note that in general, this variance cannot be computed because it depends on many population unobservables. Hence, we need to estimate it using the sample data.

# Testing Hypothesis About One Of The Regression Coefficients

## Standard Error of the *OLS* estimator

The estimator of the variance of  $\hat{\beta}_1$  replaces the unknown population values of  $\sigma_{v_i}^2$  and  $\sigma_X^2$ , by estimators of these parameters constructed from the data:

$$\hat{\sigma}_{\hat{\beta}_1}^2 = \widehat{Var}[\hat{\beta}_1] = \frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{v}_i^2}{\left[ \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right]^2}, \quad (7)$$

where  $\hat{v}_i = (X_i - \bar{X})\hat{u}_i$ .

The standard error of  $\hat{\beta}_1$  is  $SE(\hat{\beta}_1) = \sqrt{\hat{\sigma}_{\hat{\beta}_1}^2}$ .

# Testing Hypothesis About One Of The Regression Coefficients

Test  $H_0 : \beta_1 = \beta_{1,0}$  vs.  $H_1 : \beta_1 \neq \beta_{1,0}$

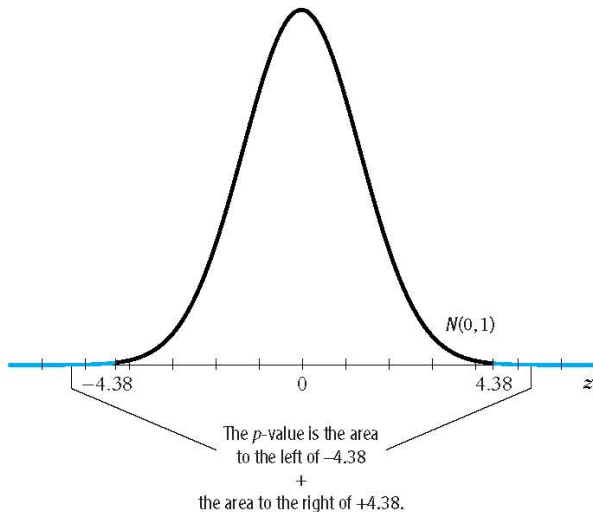
- Compute the *t*-statistic:  $t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}$ ,
- Reject  $H_0$  at the 5% significance level if  $|t^{act}| > 1.96$ .
- Compute the *p* – value for a two-sided test and reject at the  $\alpha$  significance level whenever *p* – value  $< \alpha$ .
- Recall that this procedure is based on the large sample approximation distribution of the *t*-statistic. In general,  $n = 50$  is large enough for the approximation to be excellent.

# Case Study

## Class Size and Students' Performance: The California Data

- Estimated regression line:  $\widehat{TestScore} = 698 - 2.28 \times STR$ .
- Regression software reports standard errors:  $SE(\hat{\beta}_0) = 10.4$  and  $SE(\hat{\beta}_1) = 0.52$ .
- The  $t$ -statistic testing  $\beta_{1,0} = 0$  is:  $t^{act} = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} = \frac{-2.28}{0.52} = -4.38$ .
- The 1% two-sided significance level critical value is 2.58, so we reject the null at 1% significance level.
- Alternatively, we can compute the  $p$ -value and reject at all significance levels that are smaller than the  $p$ -value.

# Case Study



# Confidence Interval For The Slope Parameter $\beta_1$

## 95% Confidence Interval

Recall that a 95% confidence interval is, equivalently

- the set of points that cannot be rejected at the 5% significance level;
- a set-valued function of the data (an interval that is a function of the data) that contains the true parameter value 95% of the time in repeated samples.

Because the *t*-statistic for  $\beta_1$  is distributed as a  $N(0, 1)$  in large samples, construction of a 95% confidence interval for  $\beta_1$  is just like the case of the sample mean

$$\text{95\% confidence interval for } \beta_1 : \left[ \hat{\beta}_1 \pm 1.96 \times SE \left( \hat{\beta}_1 \right) \right]. \quad (8)$$

# A Concise (And Conventional) Way To Report Regressions

- Put standard errors in parentheses below the estimated coefficients to which they apply

$$Test\hat{Score} = \underset{(10.4)}{698} - \underset{(0.52)}{2.28} \times STR, \quad R^2 = 0.05, SER = 18.6. \quad (9)$$

- This expression gives a lot of information
  - The estimated regression line:  $Test\hat{Score} = 698 - 2.28 \times STR$ .
  - The standard error of the estimators.
  - Measures of fit.

# OLS Regression: *STATA* Output

```
regress testscr str, robust
```

```
Regression with robust standard errors
```

```
Number of obs =      420
```

```
F(   1,   418) =    19.26
```

```
Prob > F       =    0.0000
```

```
R-squared      =    0.0512
```

```
Root MSE     =    18.581
```

```
-----+-----
```

|         |  | Robust    |           |       |       |                      |
|---------|--|-----------|-----------|-------|-------|----------------------|
| testscr |  | Coef.     | Std. Err. | t     | P> t  | [95% Conf. Interval] |
| str     |  | -2.279808 | .5194892  | -4.39 | 0.000 | -3.300945 -1.258671  |
| _cons   |  | 698.933   | 10.36436  | 67.44 | 0.000 | 678.5602 719.3057    |

```
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```

# Regression When $X$ is Binary

In many important economic applications we have binary regressors

- $X = 1$  if small class size,  $X = 0$  if not.
- $X = 1$  if female,  $X = 0$  if male.
- $X = 1$  if treated,  $X = 0$  if not treated.
- Binary regressors are sometimes called “dummy” variables.
- So far,  $\beta_1$  has been called “slope”, but that does not make sense if  $X$  is binary.
- How do we interpret regression with a binary regressor?

## Regression When $X$ is Binary

Consider the linear regression model where  $X$  is binary

- When  $X_i = 0$ ,  $Y_i = \beta_0 + u_i$ , the mean of  $Y_i$  is  $\beta_0$ , that is  $E[Y_i|X_i = 0] = \beta_0$ .
- When  $X_i = 1$ ,  $Y_i = \beta_0 + \beta_1 + u_i$ , the mean of  $Y_i$  is  $\beta_0 + \beta_1$ , that is  $E[Y_i|X_i = 1] = \beta_0 + \beta_1$ .
- So  $\beta_1 = E[Y_i|X_i = 1] - E[Y_i|X_i = 0]$  is the population difference in group means.

## Regression When $X$ is Binary

Consider the linear regression model where  $X$  is binary

$$\text{Let } D_i = \begin{cases} 1 & \text{if } STR_i < 20 \\ 0 & \text{if } STR_i \geq 20 \end{cases}$$

The *OLS* regression is:  $\widehat{TestScore} = 650 + 7.4 \times D$ .

Table a: Group Means

| Class Size                | Average Score | Std.Dev. | N   |
|---------------------------|---------------|----------|-----|
| Small ( $STR < 20$ )      | 657.4         | 19.4     | 238 |
| Large ( $STR_i \geq 20$ ) | <b>650</b>    | 17.9     | 182 |

# Regression When $X$ is Binary

## Hypothesis Testing With a Binary Regressor

- $SE(\hat{\beta}_1)$  has the usual interpretation.
- *t-statistics*, *p-values*, and confidence intervals computed as usual.
- Note that testing for  $\beta_1 = 0$  is a test of differences in means in which the null hypothesis is that both population groups have the same mean.

# Heteroskedasticity And Homoskedasticity

We need to discuss

- ① What do these terms mean?
- ② Consequences of heteroskedasticity.
- ③ Implication for computing standard errors.

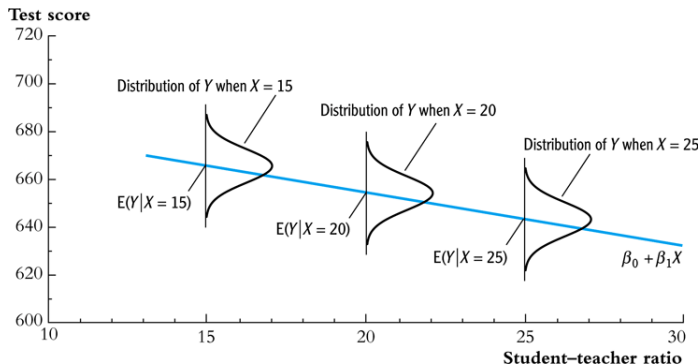
# Heteroskedasticity And Homoskedasticity

## Heteroskedasticity and Homoskedasticity

If  $\text{Var}[u|X = x]$  is constant –that is, if the variance of the conditional distribution of  $u$  given  $X$  does not depend on  $X$ – then  $u$  is said to be **homoskedastic**.

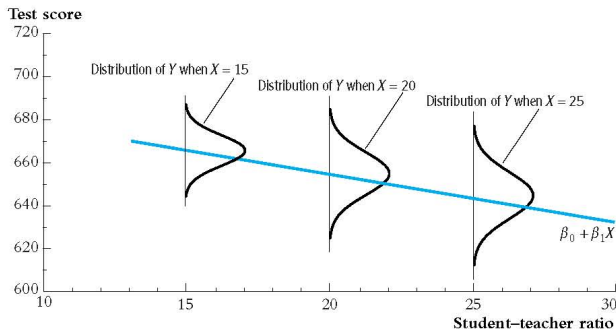
Otherwise,  $u$  is said to be **heteroskedastic**.

# Homoskedasticity in a Picture



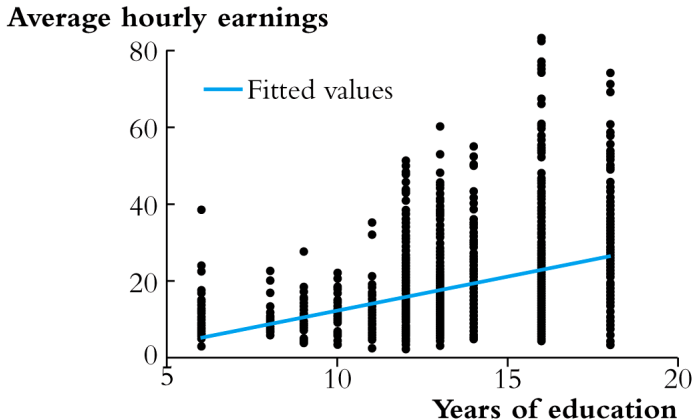
- $E[u|X] = 0$  (OLS assumption 1).
- The variance of  $u$  does not depend on the value of  $X$ ,  
 $Var(u_i|X_i = x) = \sigma_u^2$ .

# Heteroskedasticity in a Picture

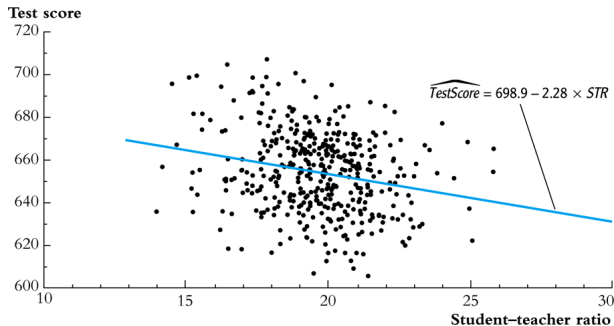


- $Var[u_i|X_i = x]$  changes with  $x$ ,  $Var[u_i|X_i = x] = \sigma_u^2(x)$
- In this particular example the variance increases in  $x$ .

## Example: Average Hourly Earnings vs. Years of Education



## Example: Test Score vs. Student-Teacher Ratio



# Heteroskedasticity And Homoskedasticity

So far we have (without saying so) assumed that  $u$  might be heteroskedastic. Recall the three *OLS* assumptions:

- 1  $E[u_i|X_i = x] = 0$ .
- 2  $(X_i, Y_i)$ ,  $i = 1, \dots, n$ , are i.i.d.
- 3 Large outliers in  $X$  and/or  $Y$  are rare.

Heteroskedasticity and homoskedasticity concern with  $\text{Var}[u|X = x]$ , thus we have implicitly allowed for heteroskedasticity.

# Heteroskedasticity And Homoskedasticity

## What if the errors are indeed homoskedastic?

- You can prove that the *OLS* estimator has the lowest variance among estimators that are linear in  $Y$ , a result called the *Gauss-Markov Theorem* to which we will return to shortly (i.e., the *OLS* estimator is efficient).
- The formula for the standard deviation of the *OLS* estimator  $\hat{\beta}_1$  simplifies to

$$\text{Var}[\hat{\beta}_1] = \sigma_{\hat{\beta}_1}^2 = \frac{\sigma_u^2}{n\sigma_X^2}, \quad (10)$$

where  $\text{Var}[u_i|X_i = x] = \sigma_u^2$ . Note that as suggested before, more spread in  $X$  means more precise estimates for  $\hat{\beta}_1$ .

## Deriving the formula for homoskedasticity-only variance

- Recall

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\text{Var}[(X_i - \mu_X)u_i]}{(\text{Var}[X_i])^2} = \frac{1}{n} \frac{\sigma_v^2}{(\sigma_X^2)^2} \quad (11)$$

- Consider the term in the numerator  $\text{Var}[(X_i - \mu_X)u_i] = E\left\{\left((X_i - \mu_X)u_i - E[(X_i - \mu_X)u_i]\right)^2\right\} = E\left\{[(X_i - \mu_X)u_i]^2\right\} = E[(X_i - \mu_X)^2 u_i^2] = E[(X_i - \mu_X)^2 \text{Var}(u_i|X_i)] = \sigma_u^2 \sigma_X^2$ , where the first equality is by the definition of variance, the second one follows because  $E[(X_i - \mu_X)u_i] = 0$  and the third one follows from the law of iterated expectations.
- substituting in (11) yields

$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \frac{\sigma_v^2}{(\sigma_X^2)^2} = \frac{1}{n} \frac{\sigma_u^2 \sigma_X^2}{(\sigma_X^2)^2} = \frac{1}{n} \frac{\sigma_u^2}{\sigma_X^2}$$

# Heteroskedasticity And Homoskedasticity

What if the errors are indeed homoskedastic?

Along with this homoskedasticity-only formula for the variance of  $\hat{\beta}_1$ , we have **homoskedasticity-only** standard errors

$$SE[\hat{\beta}_1] = \sqrt{\frac{1}{n} \frac{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}. \quad (12)$$

# Heteroskedasticity And Homoskedasticity

## Practical Implications

- Homoskedasticity-only standard errors are valid only if the errors are homoskedastic.
- The usual standard errors (to differentiate the two) are called **heteroskedasticity-robust** standard errors, because they are valid whether or not the errors are heteroskedastic.
- The main advantage of the homoskedasticity-only standard errors is that the formula is simpler. But the disadvantage is that the formula is only correct if the errors are homoskedastic.
- The *homoskedasticity-only* formula for the standard error of  $\hat{\beta}_1$  and the *heteroskedasticity-robust* formulas differ –so in general, you get different standard errors using the different formulas.

# Heteroskedasticity And Homoskedasticity

## Practical Implications

- *Homoskedasticity-only* standard errors are the default setting in regression software –sometimes the only setting (e.g. Excel). To get the general *heteroskedasticity-robust* standard errors you must override the default.
- If you do not override the default and there is in fact heteroskedasticity, your standard errors (and t-statistics and confidence intervals) will be wrong.
- Typically, homoskedasticity-only  $SE$ 's are too small (and thus, t-statistics are too large). In other words, we are more likely to reject  $H_0$ .

# OLS Regression: *STATA* Output

```
regress testscr str, robust
```

```
Regression with robust standard errors
```

```
Number of obs =      420
```

```
F(   1,   418) =    19.26
```

```
Prob > F       =    0.0000
```

```
R-squared      =    0.0512
```

```
Root MSE      =    18.581
```

```
-----+-----
```

|         |           | Robust    |       |       |                      |           |
|---------|-----------|-----------|-------|-------|----------------------|-----------|
| testscr | Coef.     | Std. Err. | t     | P> t  | [95% Conf. Interval] |           |
| str     | -2.279808 | .5194892  | -4.39 | 0.000 | -3.300945            | -1.258671 |
| _cons   | 698.933   | 10.36436  | 67.44 | 0.000 | 678.5602             | 719.3057  |

```
-----+-----
```

# Heteroskedasticity And Homoskedasticity

## Practical Implications

- If the errors are either homoskedastic or heteroskedastic and you use heteroskedastic-robust standard errors, you are OK!
- If the errors are heteroskedastic and you use the homoskedasticity-only formula for standard errors, your standard errors will be wrong (the homoskedasticity-only estimator of the variance of  $\hat{\beta}_1$  is inconsistent if there is heteroskedasticity).
- The two formulas coincide (when  $n$  is large) in the special case of homoskedasticity.

**So, you should always use heteroskedasticity-robust standard errors.**

## Additional Theoretical Foundations of *OLS*

- We have already learned a great deal about *OLS*: *OLS* estimators are unbiased and consistent; we have a formula for heteroskedasticity-robust standard errors; and we can construct confidence intervals and test statistics.
- Also, a very good reason to use *OLS* is that everyone else does –so by using it, others will understand what you are doing. In effect, *OLS* is the language of regression analysis, and if you use a different estimator, you will be speaking a different language.

## Additional Theoretical Foundations of *OLS*

- Still you may wonder; Is this a good reason to use *OLS*? Aren't there other estimators that might be better in particular, ones that might have a smaller variance?
- To answer these questions we will need to make some stronger assumptions than the three least squares assumptions already discussed.

# Additional Theoretical Foundations of *OLS*

## Extended *OLS* Assumptions

- 1  $E[u_i|X_i = x] = 0$ .
- 2  $(X_i, Y_i)$ ,  $i = 1, \dots, n$ , are i.i.d.
- 3 Large outliers in  $X$  and/or  $Y$  are rare;  $E[X^4] < \infty$  and  $E[Y^4] < \infty$ .
- 4  $\text{Var}[u_i|X_i = x] = \sigma_u^2$ , i.e.,  $u$  is homoskedastic.
- 5  $u_i$  is distributed  $N(0, \sigma_u^2)$ .

Assumptions 4 and 5 are more restrictive –so they apply to fewer cases in practice. However, if you make these assumptions, then you can prove strong results hold when these additional assumptions are true.

# Efficiency of the *OLS* Estimators

## The Gauss-Markov Theorem

Under extended *OLS* assumptions 1-4 (the basic three, plus homoskedasticity),  $\hat{\beta}_1$  has the smallest variance among all linear estimators (estimators that are linear functions of  $Y_1, \dots, Y_n$ ).

## An Additional Result

Under all five extended *OLS* assumptions –including normally distributed errors–  $\hat{\beta}_1$  has the smallest variance of all consistent estimators (linear or nonlinear functions of  $Y_1, \dots, Y_n$ ), as  $n \rightarrow \infty$ .

# Efficiency of the *OLS* Estimators

- This is a pretty amazing result –it says that, if (in addition to *OLS* 1-3) the errors are homoskedastic and normally distributed, then *OLS* is a better choice than any other consistent estimator.
- Because an estimator that is not consistent is a poor choice, this result says that the *OLS* estimator is the BEST you can do when all the five extended *OLS* assumptions hold.

# Disadvantages of *OLS* Estimators

- ① The Gauss-Markov theorem is not that compelling:
  - The condition of homoskedasticity often does not hold (homoskedasticity is special).
  - The result (assuming homoskedasticity only) is valid for linear estimators –only a small subset of estimators.
- ② The strongest optimality result requires also normally distributed errors which is not typically the case.
- ③ *OLS* estimators are more sensitive to outliers than other estimators.

# Inference Under the Extended *OLS* Assumptions

If all the extended *OLS* assumptions hold

- 1  $E[u_i|X_i = x] = 0$ .
- 2  $(X_i, Y_i)$ ,  $i = 1, \dots, n$ , are i.i.d.
- 3 Large outliers in  $X$  and/or  $Y$  are rare;  $E[X^4] < \infty$  and  $E[Y^4] < \infty$ .
- 4  $\text{Var}[u_i|X_i = x] = \sigma_u^2$ , i.e.,  $u$  is homoskedastic.
- 5  $u_i$  is distributed  $N(0, \sigma_u^2)$ .

Then,

- $\hat{\beta}_0$  and  $\hat{\beta}_1$  are both normally distributed for ANY sample size.
- The *t-statistic* has a *Student t* distribution with  $n - 2$  degrees of freedom for any sample size.

# Inference Under the Extended *OLS* Assumptions

## Practical Implication

- If  $n < 50$  and you really believe that, for your application,  $u$  is homoskedastic and normally distributed, then use the  $t_{n-2}$  instead of the  $N(0, 1)$  critical values for hypothesis tests and confidence intervals.
- In most econometric applications, there is no reason to believe that  $u$  is homoskedastic and normal –usually, there are good reasons to believe that neither assumption hold.
- Fortunately, in modern applications,  $n > 50$ , so we can rely on the large sample results based on the *CLT*, to perform hypothesis tests and construct confidence intervals using the large sample normal approximation.

## Case Study: Summary and Assessment

Suppose new teachers are hired so the student-teacher ratio falls by one student per class. What is the effect of this policy intervention (“treatment”) on test scores?. Does our regression analysis using the California data set answer this convincingly?

- Not really –districts with low  $STR$  tend to be ones with lots of other resources and higher income families, which provide kids with more learning opportunities outside school.
- This suggests that  $Corr(u_i, STR_i) \neq 0$ , so  $E[u_i|X_i] \neq 0$ .
- It seems that we have omitted some factors, or variables, from our analysis, and this has biased our results.